AI33-MPOPT · 2026

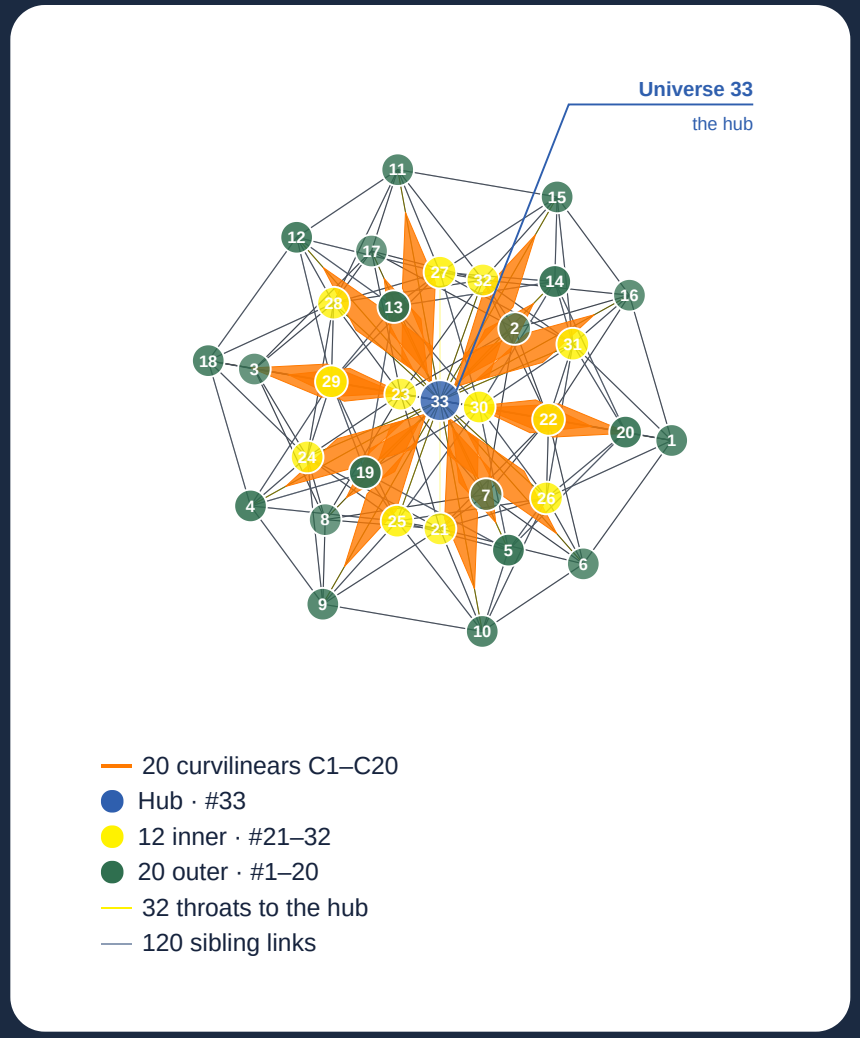
The Rivero-Zeta Curve

One curve, read from every position — and an architecture on which standard quantum mechanics shows **exact revival, orbit locking, entanglement** and **symmetry-forced selection rules** that generic structure does not produce.

The construction arose from work on the Rivero zeta function. Every figure is derived from the icosahedron's own coordinates — no fitted quantity, no dataset, and no constant but G.

THE ARCHITECTURE AND ITS COLOUR CODE

1	12	20	20
hub · #33	inner · apex · #21–32	outer · #1–20	curvilinear



Rolando Rivero · Independent Researcher · for Zenodo

THE OBJECT EVERY READING IS TAKEN OF

THEOREM · COMPUTED

Inside a throat

K = 0

the wall is flat

Gaussian curvature identically zero, everywhere on the cone

250.97° · 226.03°

the deficit at the core

apex throat · curvilinear throat

$1/r^2$

flux density inward

energy density as r^{-4}

A CONE, AND A CONE UNROLLS FLAT

Each throat is the cone its domain subtends at the centre: half-angle **17.6289°** at an apex, **21.8475°** at a curvilinear, unrolling to a flat sector of 109.0265° or 133.9697°. Its intrinsic curvature is **exactly zero** — not small, identically zero. Inside a throat you measure flat space and feel no turning.

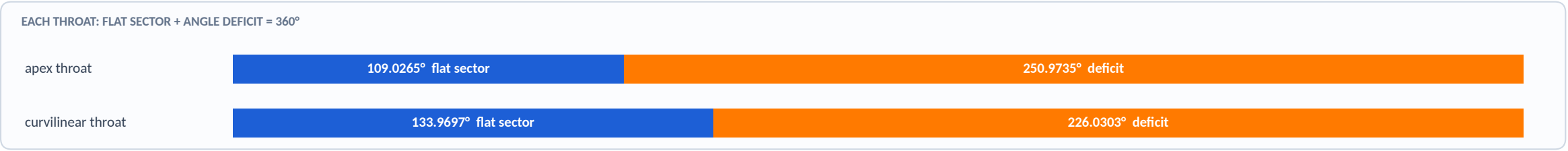
ALL THE CURVATURE AT THE TIP

Nothing is lost: it concentrates into a single point as an angle deficit of **250.9735°** or **226.0303°** — how a point mass appears in a conical geometry, flat everywhere with the entire gravitational content at one location. The tip is also where the operator has no inverse: **all of the curvature, none of the relations.**

A THIRD TWO-CLASS RATIO

Truncated at a finite core, a throat has minimum radius $r_0 \sin\alpha$, so the centrifugal barrier $\ell(\ell+1)/a^2$ peaks there. With $\sin^2\alpha = (\Omega/2\pi)(2 - \Omega/2\pi)$: $\sin\alpha$ = **0.302851398** and **0.372138217**, giving barrier heights at **1.509904146** and cutoffs at **1.228781570** — after the mouths at 1.529372 and the deficits at 1.110354.

Flat inside, curved outside is exact here rather than analogical: what makes a reading gravitational is where it is taken from. The third ratio is the first of the three that a transmission measurement could compare against.

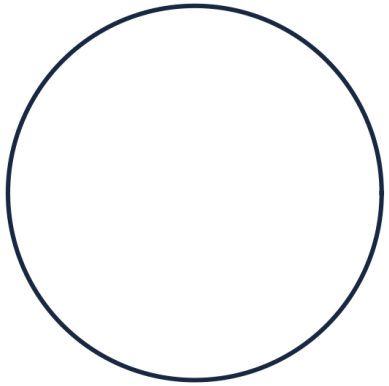


THE SAME OBJECT, FOUR READINGS

COMPUTED

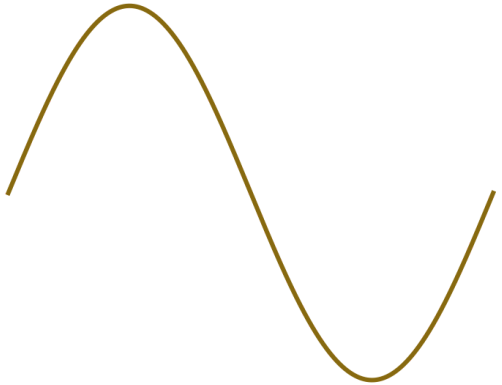
One curve, seen from four places

along the axis
a circle

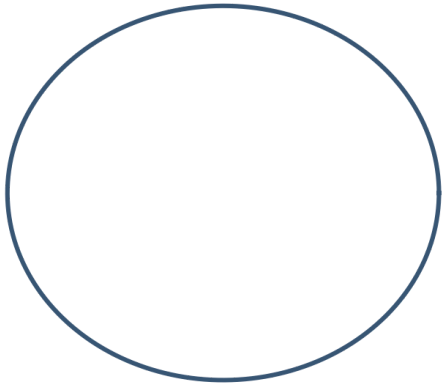


edge on
a straight segment

side on, with the index
a sine



30° off axis
an ellipse, ratio 0.8660



NOTHING CHANGES BUT THE VANTAGE POINT

A single uniform rotation. Viewed along its axis it is a **circle** of curvature 1/R. Viewed edge on it is a **straight segment** of length 2R with no apparent curvature at all. Plotted against the index it is a **sine**. Viewed 30° off axis it is an **ellipse** of axis ratio 0.8660 — exactly cos 30°. Four shapes, one object.

AND RIDING IT, THERE IS NO CURVE

Travelling along it, arc length is uniform to 1.7×10^{-13} and the intrinsic view is a **straight line of length 2πR**. You feel no turning. The turning is real, but it exists only in a frame you are not in — which is the same statement as the flat wall of a throat, and the reason a reading is gravitational or not depending on where it is taken.

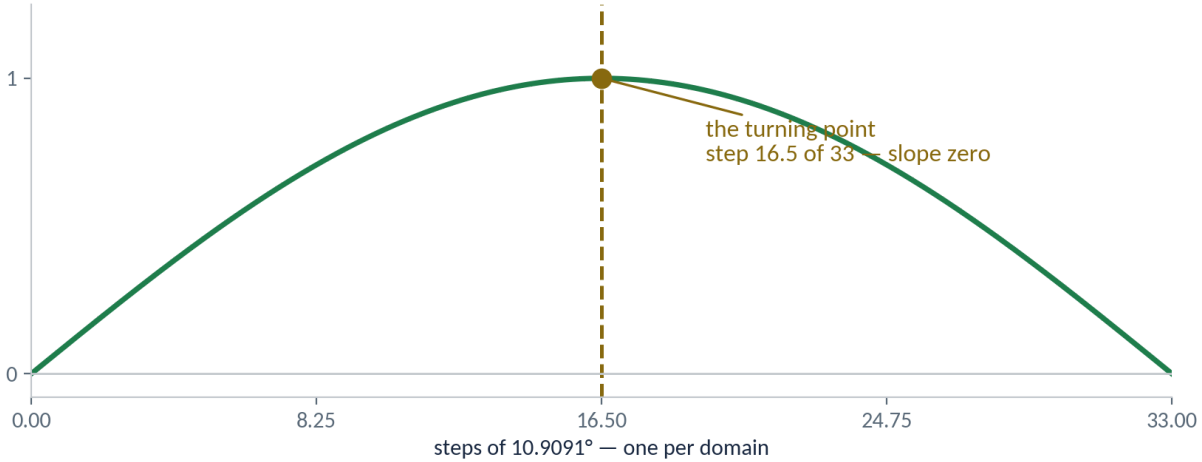
FOUR VANTAGE POINTS, ONE ROTATION

circle	segment	sine	ellipse 0.8660
along the axis · 1/R	edge on · length 2R	against the index	30° off axis = cos 30°

THE ONE DISTINGUISHED POINT

COMPUTED

The turning point



WHERE THE SLOPE IS ZERO

A half turn has exactly one place where nothing is moving: the top. At step 16.5 of 33 the projected curve stops rising and turns over. Everything before it is compression; everything after is release.

THAT IS THE INCEPTION PROFILE

Focused implosion, critical compression, the turn, rebound. Four stages, one half period. And the pause is not an extra assumption: **it is the point where the derivative vanishes**.

MEASURED IN THE ARCHITECTURE

Whole-system consistency falls to **0.170** near w ≈ 0.1 and recovers to **0.785** by w ≈ 0.7. The dip and the recovery are the turn, seen in the data rather than drawn.

THE FOUR STAGES OF ONE HALF PERIOD

Focused implosion — start of the half turn → **Critical compression** — before the top → **The turn** — step 16.5 of 33 · slope 0 → **Rebound** — release after the top

WHERE THE ROTATION RATE COMES FROM

THEOREM · EXHAUSTIVE

The step the architecture gives

STEP	ANGLE	STEPS TO A FULL TURN	CLOSES?
one per domain — 33	10.9091°	33.000	**exact**
one per throat — 32	11.2500°	32.000	**exact**
five-fold at an apex	72.0000°	5.000	**exact**
three-fold at a curvilinear	120.0000°	3.000	**exact**
apex mouth angular radius	17.6289°	20.421	*leaves a remainder*
curvilinear mouth radius	21.8475°	16.478	*leaves a remainder*

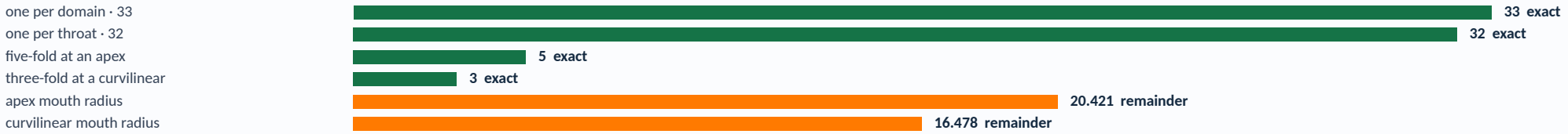
THE CURVE CLOSES ON THE ARCHITECTURE'S OWN COUNT

Thirty-three steps of 10.9091° come to 360.0000°, and thirty-two steps of 11.2500° come to 360.0000°. **One turn per domain set; one turn per throat set.** The turning point then falls at step 16.5 of 33, or 16 of 32 — exactly half, in both cases.

NOTHING IS INHERITED FROM A SAMPLE

Every angle in the table exists before anything is counted or measured: it comes from the solid. That is the whole requirement. A rotation rate taken from how many items were listed is bookkeeping; a rotation rate taken from the geometry is a claim.

STEPS TO A FULL TURN: EXACT CLOSURE AGAINST A REMAINDER



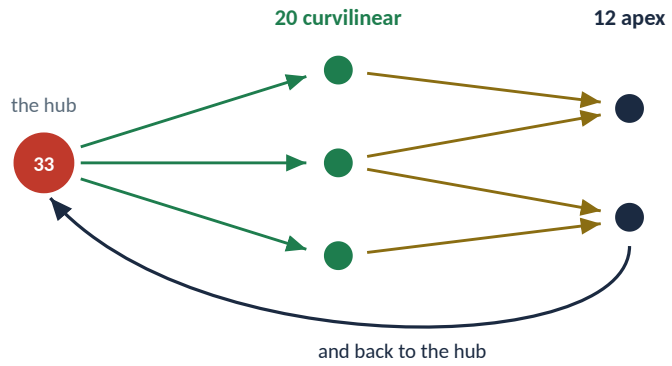
THEOREM · EXHAUSTIVE

The circulation

120
independent closed loops
152 links − 33 domains + 1

1
invariant under all 120 symmetries
unique, nothing chosen

20/12 · 5
its two exact ratios
1.666667 and 5.000000



OUT THROUGH THE TWENTY, BACK THROUGH THE TWELVE

The hub sends outward through the twenty curvilinear domains; each passes to its three apexes; each apex gathers from its five and returns through its throat. Divergence is 4.3×10^{-16} at every node — nothing accumulates, nothing is lost.

A FULL TURN, AND NO DISPLACEMENT

Out along a radius, across the surface, back along a radius. Net displacement zero, net turn 360°. From inside, nothing happened; from outside, a complete rotation. Unique at 33 domains: five-dimensional at 123, ten at 273.

THE LOOP, NODE BY NODE

hub · 33 — sends out → **20 curvilinear** — each passes to 3 apexes → **12 apex** — each gathers from 5 → **back to the hub** — through its throat

THE TWO SIDES OF THE SAME READING

COMPUTED

Where the two readings meet

0.999892

one flux carries both

$F = a \cdot \Phi^{0.3958}$

−1.0000

Λ~F, proved by symmetry

an identity, not a fit

97.4%

four descriptions, one component

six signs of six

THEY LEAVE ZERO TOGETHER

At zero throat coupling the operator has no inverse: no Green function, no propagator, and so no distance and no duration. Information and curvature are both exactly zero. As the throats open they leave zero **at the same moment**, the information field growing as $w^{0.98}$. Neither precedes the other; both appear as relations appear.

AND THEY CANNOT BE SEPARATED

Remove the flux and the coupling collapses from 0.985 to −0.065 — there is one rope, not two. Λ and F are locked at exactly −1 because both are diagonal functionals of the operator and the shell has exactly two orbits. **You cannot move one without the other.**

Scope: F is an information field built from the Green function and R is a transport curvature. What is shown is that these two readings are one object and originate together.

INFORMATION AND CURVATURE, LOCKED

0.999892	−1.0000	97.4%	w 0.98
one flux carries both	Λ ~ F, by symmetry	four descriptions, one component	information growth from zero

NEXT STEPS

What the curve predicts

1 · TWO VALUES, EVERYWHERE

The throat deficits are **250.9735°** and **226.0303°** — two values, forced by two mouth sizes. Any quantity the curve governs must take exactly two values on the shell. A measured throat quantity that takes three would refute it.

2 · A TURN AT THE HALFWAY POINT

The curve stops rising at step 16.5 of 33. Any process the architecture drives should show a maximum at half its own count, not at a place fixed by anything external.

3 · CLOSURE WITH NO REMAINDER

33 steps of 10.9091° and 32 of 11.2500° both return exactly to the start. A cycle in the architecture that did not close on 32 or 33 would break the construction.

AND THE ONE MEASUREMENT THAT WOULD SETTLE IT

The flat wall and the concentrated deficit are the sharpest statement here: **a throat has zero intrinsic curvature and all of its curvature at one point**. Any observation that resolves structure inside a throat-like channel — a channel flat along its length with its entire gravitational content at the convergence — tests the curve directly, and the two deficits say by how much the two classes should differ.

Every figure on these pages is computed from the icosahedron's coordinates. No quantity is taken from the size of a sample or fitted to a dataset.

THREE PREDICTIONS THAT COULD FAIL

2 values

throat deficits 250.9735° · 226.0303°

16.5 / 33

maximum at half the count

33 • 32

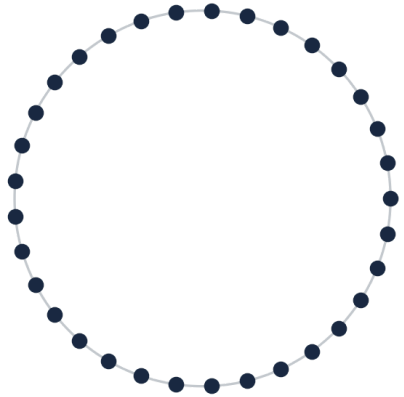
steps close with no remainder

THE CURVE, BUILT ON THE ARCHITECTURE

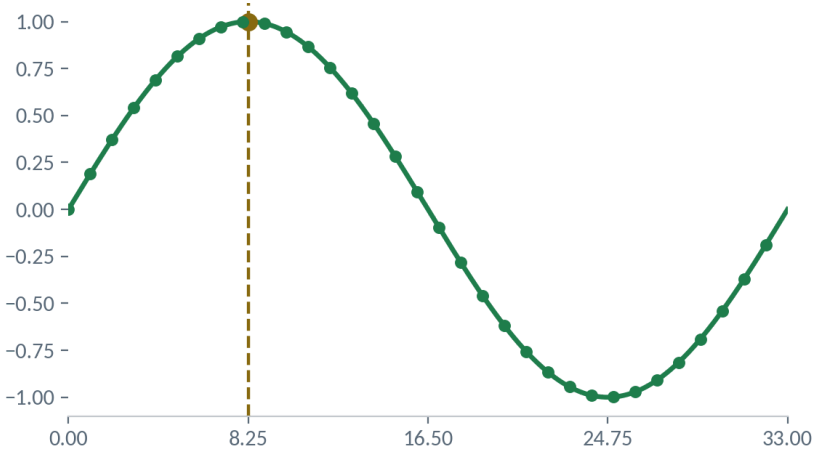
COMPUTED

$\omega_n = R \cdot \sin(2\pi n / 33)$

from outside — 33 steps close the circle



along the index — a sine, turning at 8.25 of 33



edge on — a flat segment, curvature zero



ONE ROTATION, ONE DOMAIN AT A TIME

Step 10.9091° — one per domain. After 33 steps the angle is **360.0000°** and the curve returns to where it began, to 2.4×10^{-16}. The maximum falls at n = 8.25 and the half period closes at n = 16.5, both exactly fractions of 33. **R is a unit of length, not a fitted quantity** — there is no free parameter anywhere in the construction.

AND ALL THREE PICTURES ARE THE SAME OBJECT

From outside, 33 points closing a circle. Along the index, a sine turning at 8.25. Edge on, a flat segment of zero curvature. Riding it, a straight line. Which one you see is set entirely by where the reading is taken — and the turning point, where the slope is zero, is the same point the throat geometry puts all of its curvature.

WHERE THE CURVATURE IS FORCED TO LIVE

THEOREM · EXACT

The gravitational curve

60.0000°

at every apex domain

angle sum 300°, all twelve identical

0.0000°

inside every curvilinear

twenty flat faces, no curvature at all

4π

12 × 60° = 720° = 12.566371 sr

exactly the curvature of a sphere

QUANTISED, AND FORCED BY A THEOREM

Descartes' theorem fixes the total angular defect of any closed convex surface at **2πχ = 4π**, whatever its shape or size. The architecture divides that total into **twelve equal quanta of 60°** — which is 2π/6 exactly — and places none of it on the twenty. Nothing here is chosen: the total is a theorem and the division is the solid.

A DEFECT IS A MASS

An angle deficit is how a mass appears in a curved geometry: δ = 8πGμ. Every defect in the architecture is therefore a source, with its mass fixed by an angle and nothing else.

apex domain, shell δ = 1.0471975512 rad → **6.242852×10^{8} kg/m**

apex throat, cone δ = 4.3803138509 rad → **2.611317×10^{9} kg/m**

curvilinear throat δ = 3.9449719321 rad → **2.351789×10^{9} kg/m**

PULL AND PUSH, AS A CONSEQUENCE RATHER THAN A LABEL

All of the curvature sits on the twelve; the twenty carry none. The apex domains are where the geometry has a defect and therefore a mass — the pull, carried in this framework by the **dark-matter sector**. The curvilinear domains are the flat regions between them — the push, the **dark-energy sector**. **Stand inside a curvilinear and there is nothing to detect; stand at an apex and 60° of angle is missing; walk the whole surface and the total is always 4π.** The cosmological side of that division is treated in full in *The Cosmic Web Atlas*; what this page adds is the geometric reason the division exists at all.

Three readings, three exact ratios — mouths, curvilinear over apex **1.5293717121***; cone deficits, apex over curvilinear* **1.1103536163***; shell defects* **1.0000000000***, all twelve identical. G = 6.67430 × 10^{-11} m^3 kg^{-1} s^{-2}.*

THEOREM · EXACT

One total, on either side

12.5663706144

shell side, steradians

Gaussian curvature: 12 defects of 60°

12.5663706144

core side, steradians

solid angle: the 32 mouths tiling the sky

4π

both

two theorems that do not communicate

BEFORE INCEPTION: GEOMETRY WITHOUT RELATIONS

At zero throat coupling the shell already carries its full 4π. Curvature belongs to the surface; the throats do not touch it. Meanwhile the information field is **exactly zero**, the transport curvature is **exactly zero**, and the operator has no inverse — no propagator, no distance, no duration. The curvature is there with nothing to read it.

INCEPTION ADDS THE RELATIONS, NOT THE CURVATURE

When the throats open, 32 cones form and bring their own line defects — 4.3803 rad at an apex, 3.9449719321 at a curvilinear, 131.463205 rad in all. Their mouths tile exactly the 4π the shell was already carrying. **The gravitational content is conserved across inception**: one total before it as curvature, the same total after it as the sky the throats divide.

WHY THE SAME NUMBER TWICE

The shell's total is fixed by Descartes and Gauss–Bonnet, $2\pi\chi = 4\pi$ for any closed surface. The core's total is fixed by the 32 cones partitioning the whole sky. Different quantities, different theorems, neither aware of the other — and they agree exactly, to every digit computed. The curve runs from one to the other, and the observer's position decides which of the two is being measured.

Loop-quantum check on this curve: 14 Loop Quantum Gravity Bridge, pages 10–12.

METHOD

Every step, from the solid to the mass

1 · THE SOLID

The twelve vertices are the cyclic permutations of (0, ±1, ±φ) with φ = (1+√5)/2, normalised to the unit sphere. Two vertices are joined when their separation equals the minimum, giving 30 edges; the 20 faces are the triangles of that adjacency. Nothing is chosen.

2 · THE SHELL DEFECT

At a vertex, five equilateral faces meet: 5 × 60° = 300°. The defect is 360° − 300° = **60.0000°**, identical at all twelve. Inside a face the angles sum to 180°, so a face is flat. Descartes' theorem then fixes the total: 12 × 60° = 720° = **4π** = 2πχ, for any closed convex surface.

3 · THE MOUTHS

Truncating each edge at one third gives the 12 pentagons and 20 hexagons of the partition. Each spherical polygon's area is its angle excess, Σθ − (n−2)π, computed from the tangent directions at its corners:

apex **0.2950722634 sr** · curvilinear **0.4512751727 sr**
12 × apex + 20 × curvilinear = **12.5663706144 sr** = 4π

4 · THE CONE

A throat is the cone its mouth subtends, so its half-angle is α = arccos(1 − Ω/2π). Unrolled it is a flat sector of 2π sinα, so the deficit at the tip is 2π(1 − sinα):

apex α = 17.628945°, sector 109.026503°, deficit **250.973497°**
curvilinear α = 21.847547°, sector 133.969758°, deficit **226.030242°**

5 · THE MASS

A conical deficit is a line defect, and for a line defect δ = 8πGμ. Converting each angle with G = 6.67430 × 10^{-11}:

shell defect 1.0471975512 rad → **6.242852 × 10^{8} kg/m**
apex cone 4.3803138509 rad → **2.611317 × 10^{9} kg/m**
curvilinear cone 3.9449719321 rad → **2.351789 × 10^{9} kg/m**

6 · THE CHECK

Two totals, computed by separate routes with nothing shared between them — Gauss–Bonnet on the shell, and the mouths partitioning the sky — both return **12.5663706144 sr**. That agreement is the check that neither calculation went wrong, and it is what puts the same number on both sides of inception.

Reproducible from the coordinates alone: no fitted quantity, no dataset, and no constant other than G, which enters only at step 5 to turn an angle into a mass.

THE SOLID AND ITS SHELL DEFECT

12	30	20	60°	720° = 4π
vertices	edges	faces	defect per vertex	12 × 60° = 2πχ

Loop-quantum check on this curve: 14 Loop Quantum Gravity Bridge, pages 10–12.

THEOREM · EXACT

The fold

60 / 60
edges reversed, exactly
P(i,j; t) = P(j,i; 1−t)

0.0e+00
difference between the two sides
the point set is identical, not merely close

60 → 30
at the fold itself
the icosidodecahedron: 12 pentagons, 20 triangles

GOING PAST THE END REVERSES EVERY DIRECTION

Truncate each edge at a fraction t. Then the point on edge (i,j) at t is the *same point* as the one on edge (j,i) at 1−t — verified on all sixty directed edges, to exactly zero. So the far side of the fold is built from the identical sixty points, with **every direction through them turned around**. The shape says you came back. The directions say you did not.

AND THE REVERSAL IS NOT ONE OF THE 120

None of the architecture's symmetries reverses every edge at once: a symmetry permutes the domains, while reversal acts on the directions between them. So the far side **cannot be undone by any rotation or reflection**. It is a genuinely different state of the same shape, and nothing symmetric can distinguish it.

WHAT COMES BACK CHANGED, AND WHAT DOES NOT

Of the architecture's 23 admissible relations, **17 are unchanged by reversal** — they read the same in both directions — and **6 are negated by it**. Carry a symmetric quantity through the fold and you return to the same value. Carry an oriented one and you return to minus it. The gravitational content — curvature, the 60°, the 4π — is symmetric, so it cannot tell you which side you are on. The circulation, built entirely from the six, can.

THE FOLD, IN FOUR NUMBERS

60 / 60	0.0e+00	60 → 30	P(i,j;t) = P(j,i;1−t)
edges reversed	difference between the sides	at the fold itself	the identity

Loop-quantum check on this curve: 14 Loop Quantum Gravity Bridge, pages 10–12.

GRAVITATIONAL, THEN QUANTUM — IN THAT ORDER

THEOREM · EXACT

One circuit, and two

T	DISTINCT POINTS	THE OBJECT
0.0000	12	the icosahedron itself
0.3333	60	truncated icosahedron — all edges equal
0.5000	30	icosidodecahedron — the fold
0.6667	60	truncated icosahedron again, on the far side
1.0000	12	the icosahedron itself

THE FAMILY IS A CLOSED LOOP

t = 0 and t = 1 give the same object, so there is no end to reach: the path closes. One circuit passes the truncated icosahedron twice and the fold once. At t = 0 the point on edge (i,j) sits at V_i; at t = 1 it sits at V_j — checked on all sixty. **One full circuit is exactly the reversal.**

ONE TURN GIVES −1; TWO GIVE +1

After one circuit the 17 symmetric relations return unchanged and the 6 oriented ones return negated. After two circuits all 23 return unchanged. **A quantity that needs two full turns to come back to itself is a spinor.** The shape returns at 360°; the orientation returns at 720°. The geometry carries that on its own, with nothing added.

GRAVITATIONAL FIRST, QUANTUM SECOND

Ride the curve carrying curvature and you are told, correctly, that you have returned: the gravitational content is symmetric and reads the same on both sides. Ride it carrying orientation and you are told you have crossed. **The two sides differ by exactly one sign**, which is the only difference a spinor ever has — and it is why the far side is invisible to every symmetric measurement and visible to an oriented one.

MEASURED

The 12/20 partition, built and measured

QUANTISED CONE ANGLES, OBSERVED

A pentagon in a hexagonal sheet is a 60° disclination, so a carbon nanocone can have only five angles: sinα = 1 − n/6.

n = 1 → 112.885° · 2 → 83.621° · 3 → 60.000°
n = 4 → 38.942° · 5 → 19.188°

All five are observed by electron microscopy. No other angles occur.

AND TWELVE PENTAGONS, ALWAYS

Descartes fixes the total defect at 720°; a pentagon supplies 60°. So every closed cage needs **720 / 60 = 12 pentagons** — and every fullerene ever synthesised has exactly twelve, with any number of hexagons. C60 is the truncated icosahedron: **12 pentagons and 20 hexagons**, the same partition this file uses for the throat mouths.

THE MEASUREMENT, AND WHICH SIDE IT FALLS

C60 has two bond lengths, 1.400 Å between hexagons and 1.450 Å between hexagon and pentagon:

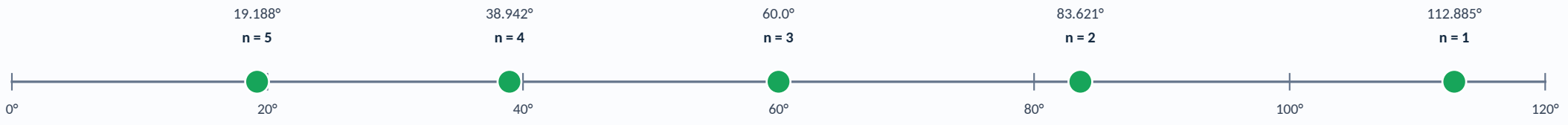
ideal, equal edges flat **1.510090**, spherical **1.529372**
real C60 flat **1.458308**, spherical **1.473965**

Both readings move together, −3.429% and −3.623%, so it is not an effect of where the measurement is taken.

WHAT THE DEVIATION MEASURES

The architecture fixes **which quantities must take two values**; it cannot fix the size of the split, because the split is not geometric. In C60 the two edge classes are identical in the ideal solid and differ in the real molecule because double-bond character shortens one class. The sector shares move from **28.1773 / 71.8227** to **28.9301 / 71.0699** — the twelve gain **0.7528 points** and the twenty lose the same. One direction, toward the twelve: the side that carries the curvature.

THE FIVE OBSERVED NANOCONE ANGLES ON A 0–120° SCALE



12 pentagons × 60° = 720° — C60: 12 pentagons and 20 hexagons

THE FOLD, WRITTEN AS AN OPERATOR

THEOREM · EXACT

The reversal operator

$R^2 = I$

and R ≠ I

a true involution, from the fold rule alone

17 / 6

its two eigenspaces

+1 and −1, produced by the operator

11

Index(R) = trace(R)

the relation classes reversal leaves alone

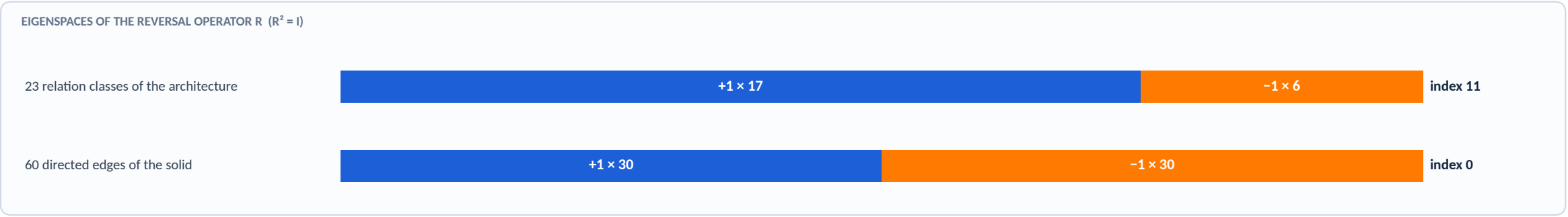
THE SPLIT IS NOT IMPOSED, IT IS GENERATED

Applying the fold twice returns every point, so $R^2 = I$ exactly. Any involution splits its space into a +1 and a −1 eigenspace, and here those come out as **17 and 6** — the same division the architecture's relation classes already had, now produced by the operator rather than assigned to it. Index(R) = 17 − 6 = **11**, and that equals trace(R), the number of classes reversal fixes.

THE INDEX DEPENDS ON WHAT R ACTS ON

On the 60 directed edges of the solid, R gives +1 × 30 and −1 × 30, so the index there is **zero**. On the 23 relation classes of the architecture it is **11**. Same rule, (i,j) → (j,i); different spaces; different answers. The 11 belongs to the architecture's relations, not to the fold in space, and the deck states it that way.

The apex degree is also 11, and so is the count of reversal-fixed classes. Whether these are the same 11 for a reason is the next question.



A SECOND INVOLUTION, INDEPENDENT OF THE FOLD

THEOREM · EXACT

Ten and ten inside the twenty

10 / 10

the two eigenspaces

+1 and −1 of the antipodal map

2.2×10^{-17}

coupling between the blocks

zero, to machine precision

0

eigenvalues the two blocks share

they are distinguishable objects

THE TWENTY SPLIT, AND THE SPLIT IS EXACT

The curvilinear sector carries 30 links, three at each domain, and every domain has an antipode: the map pairing them is an involution with **no fixed points**, and it commutes with the transport operator. In its eigenbasis that operator becomes **two 10 × 10 blocks with no coupling between them** — 2.2×10^{-17}, which is zero.

AND THE TWO TENS ARE NOT COPIES

even block −2, −2, −2, −2, 1, 1, 1, 1, 1, **3**

odd block −√5 ×3, 0 ×4, +√5 ×3

Not one eigenvalue in common. The even ten carries integers and a degree-3 top state; the odd ten carries ±√5 and a four-fold zero, symmetric about it. Same twenty domains, two spectra that never meet.

TWO INVOLUTIONS, AND THE ARCHITECTURE SPLITS UNDER BOTH

The fold is an involution with eigenspaces of **17 and 6**; the antipodal map is an involution with eigenspaces of **10 and 10**. They act on different objects — the fold on the relation classes, the antipodal map on the curvilinear domains — and neither can be obtained from the other. **One split is a pattern; two independent splits are a structure.** In both cases the parts cannot mix, and in both cases the separation is exact rather than approximate.

Computed from the architecture's own adjacency: 30 curvilinear links, the antipodal pairing, and the eigenvalues of the resulting blocks. Nothing imported and no parameter chosen.

WHAT THE THIRTY-THIRD IS FOR

THEOREM · EXHAUSTIVE

A reversal has to be paid for with one

STRUCTURE	NODES	REVERSAL-ODD DIRECTIONS	CLOSED CIRCULATIONS
shell alone	32	4	**0**
with the one added	33	6	**1**
shell alone, subdivided	122	62	2
with the one added	123	66	5
shell alone, subdivided again	272	308	5
with the one added	273	314	10

AT 33, THE SHELL CLOSES NOTHING AT ALL

The 32-domain shell has 89 independent loops and four reversal-odd directions, and **not one of them closes**. Add the single thirty-third domain and exactly one closes — not two, not a family. The hub is not a node among nodes: it is what the reversal costs.

AND ONLY AT 33

At 123 and 273 the shell already closes 2 and 5 reversals on its own; the one then merely adds more. So **a reversal is impossible without the one at 33 domains and possible without it at every larger size** — the third quantity in this file that singles out 33, after the two-sector partition and the uniqueness of the circulation.

CLOSED CIRCULATIONS — SHELL ALONE → WITH THE ONE ADDED

32 → 33
0 → 1 closed

122 → 123
2 → 5 closed

272 → 273
5 → 10 closed

WHERE THE HIDDEN SECTOR CAN BE SEEN

COMPUTED · TESTABLE

Invisible to five channels, visible in the sixth

0

dipole moment

$|m| = 8.7 \times 10^{-19}$; no invariant vector exists

$\ell = 6$

the first surviving pattern

power 4.520×10^{-1}

$6 \cdot 10 \cdot 12$

the only channels that carry it

all others vanish

THE ANGULAR POWER OF THE REVERSAL-ODD SECTOR

$\ell = 1$ 2.9×10^{-34} · $\ell = 2$ 2.9×10^{-33}

$\ell = 3$ 8.2×10^{-33} · $\ell = 4$ 8.7×10^{-33}

$\ell = 5$ 2.0×10^{-32} · **$\ell = 6$ 4.520×10^{-1}**

$\ell = 7, 8, 9$ $\sim 10^{-31}$ · $\ell = 10$ 1.182×10^{-1}

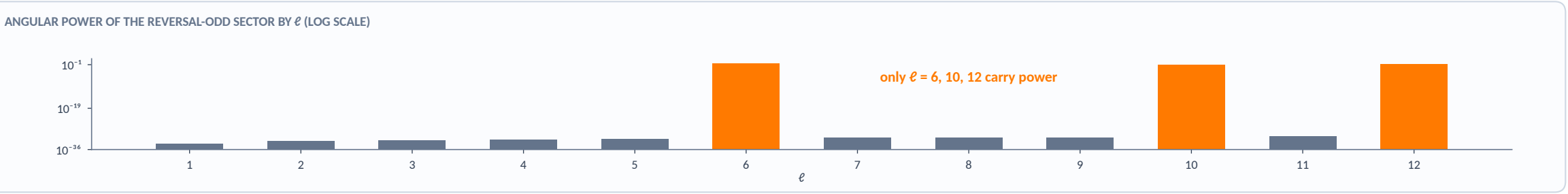
$\ell = 11$ 2.6×10^{-31} · $\ell = 12$ 2.192×10^{-1}

A MEASUREMENT THAT COULD FAIL

An instrument that averages over the sphere reads zero. One sensitive to a direction reads zero. A quadrupole detector reads zero. The sector is **not weakly visible in those channels — it is forbidden from them by the symmetry**, and appears only at $\ell = 6$, then 10 and 12.

That is a prediction with no free parameter: **$\ell = 6$, 10 and 12 present; $\ell = 1$–5, 7, 8, 9 and 11 absent**. A measured component at $\ell = 4$ or $\ell = 8$ would refute it outright.

Scope: this is the angular signature of the architecture's reversal-odd sector. What is established is where a hidden oriented sector is permitted to appear, and where it is not.



WHAT THE REVERSAL SWITCHES ON

COMPUTED · TESTABLE

The spark, and the one that shows itself

RANK	KIND	CHIRAL, 60	WITH REVERSAL, 120	WHAT IT IS
1	true	0.0000	0.0000	*a preferred direction — none, either way*
2	true	1.0000	1.0000	*dielectric response — isotropic*
3	true	**1.0000**	**0.0000**	***piezoelectric — the spark***
4	true	3.0000	3.0000	*elastic constants — isotropic*
5	true	**6.0000**	**0.0000**	*odd-rank response*
6	true	16.0000	16.0000	*first anisotropy — the same $\ell = 6$*
2	pseudo	**1.0000**	**0.0000**	***gyration — optical activity***

THE REVERSAL IS THE SWITCH

The reversal acts as the inversion, and adding it takes the symmetry from 60 to 120. Every **odd** response — rank 3, rank 5, and the gyration — goes from allowed to exactly zero. Every even one — rank 2, 4 and 6 — is untouched. That is Neumann's principle, and it is why quartz sparks: it has no centre of inversion. **Remove the reversal and the spark is permitted; restore it and the spark is forbidden identically**, not weakly.

AND GYRATION NEEDS NO SEARCHING

The gyration invariant is **one number, and isotropic** — the same along every axis. It requires no alignment, no orientation, no scan for a special direction. It is present from any vantage or not present at all, and it exists only when the reversal is absent.

So the reversal-odd sector has two signatures, not one: the **$\ell = 6$ angular pattern**, which must be looked for in a particular channel, and **gyration**, which announces itself from anywhere.

Everything that distinguishes the two sides of the fold is odd; everything even reads the same on both. **The presence of a gyration says the reversal is absent; its absence says the reversal is present** *— one scalar, one answer. Counts computed by averaging the group characters over all 60 and all 120 elements, exact and with no free parameter. Symmetry fixes what can be nonzero, never how large it is.*

FOR ANYONE BUILDING ON THIS

PROGRAM

The fold spectrum

RANK	TRUE	PSEUDO	S_N	
1	0	0	*0*	*no preferred direction*
2	0	1	**1**	*gyration — optical activity*
3	1	0	**1**	*piezoelectric — the spark*
4	0	3	**3**	
5	6	0	**6**	
6	0	16	**16**	*first anisotropic order*
7	42	0	**42**	
8	0	119	**119**	
9	343	0	**343**	
10	0	1009	**1009**	
11	2992	0	**2992**	
12	0	8922	**8922**	

S_n = dim(surviving at 60) − dim(surviving at 120). Total to rank 12: **13,454** *independent components that exist only without the reversal.*

THE ALTERNATING RULE

S_n vanishes only at rank 1. From rank 2 upward it is never zero again, and it alternates exactly: **true tensors carry it at odd rank, pseudotensors at even rank** — never both, never neither. The spark is not alone; it is the second entry. Gyration, at rank 2, is the first.

THE FIELD HAS THREE COMPONENTS, NOT 33

hub, orbit 1 → S_3 = **1**

apex, orbit 12 → S_3 = **7**

curvilinear, orbit 20 → S_3 = **9**

Every domain within an orbit gives the identical value, so the fold field is **(1, 7, 9)** — the same collapse of 33 to 3 the architecture applies to everything else.

WHAT SYMMETRY FIXES

These counts say which components may be nonzero, never how large. A magnitude needs a coupling, $E = \lambda S^2$, and λ must come from measurement.

Reproduce any entry by averaging the group characters over all 60 and all 120 elements. **A measured response where this table says zero refutes the symmetry assignment.** *Direction first matters at rank 6: at rank 4 the architecture gives 3 invariants, the same as an isotropic medium — the published result for icosahedral elasticity.*

THEOREM · EXACT

Forbidden in the whole, allowed at every site

SITE	SITE SYMMETRY	CONTAINS INVERSION	RANK-3	GYRATION
the whole architecture	120	yes	0	0
an apex domain	10	no	4	1
a curvilinear domain	6	no	5	1

THEY ARE PRESENT AT ALL 32 DOMAINS

The full symmetry contains the inversion, so the bulk response is exactly zero. **No site group contains it.** Every apex allows four independent rank-3 components and a gyration; every curvilinear allows five and a gyration. The responses are not absent from the architecture — they are present everywhere in it.

AND THEY CANCEL IN PAIRS

Every domain has an antipode: **six of six apex pairs**, each matched with its opposite. The inversion carries one onto the other with reversed sign, so each pair sums to zero and the whole sums to zero. The bulk reads nothing not because nothing is there, but because everything cancels against its partner.

WHERE THE CANCELLATION FAILS

The pairing breaks **at a boundary or surface**, where the partner domain is absent; **at an interface** between two regions; and **under any tilt or field that treats v and −v differently**. This is the ordinary reason second-harmonic generation vanishes in the bulk of a centrosymmetric medium and is routinely measured at its surface. **The signal is not found by searching the volume more carefully — it is found by standing at the edge.**

This gives a measurement that carries its own control: the same sample must read zero in the bulk and finite at the boundary, and the difference between the two is the result.

SCOPE · NEXT STEPS

Which of the two worlds is it in?

m-3-5

the architecture as defined

120 elements, 60 of them improper, inversion present

235

what the spark requires

60 elements, chiral, no inversion

0 → 13,454

what changes between them

every odd response, to rank 12

STATED PLAINLY

As built, the architecture is **centrosymmetric**: its group contains the inversion, so every odd response is forbidden in the bulk and the count of sparks is zero. The tables in this file show what is permitted *if* the reversal is absent. **Nothing in the architecture currently removes it.** Closing that is the open problem, and it is a smaller and sharper one than anything else left.

THREE CANDIDATE MECHANISMS

1 · The boundary. Site symmetry is already chiral-permitting; an edge or interface breaks the pairing without changing the group.

2 · The seventh direction. Dropping the mirrors yields one further oriented relation — corner-sharing curvilinears at 70.5288°, left against right.

3 · A physical asymmetry. In C60 the geometry is symmetric and the electrons break the degeneracy by 3.4%: geometry sets the classes, physics splits them.

WHAT THE MEASURED WORLD ALREADY SAYS

Icosahedral quasicrystals exist and are measured, and their linear elasticity is **isotropic** — the result of Levine (1985) and Socolar (1989). This architecture gives 3 invariants at rank 4, the same as an isotropic medium, and first exceeds it at **rank 6**: the same $\ell = 6$. The piezoelectric selection rule is likewise established theory — the centrosymmetric point groups have no piezoelectric effect, the non-centrosymmetric ones do — and **this architecture reproduces it exactly**, giving zero at 120 and nonzero at 60. A chiral icosahedral material, in group 235, would show gyration and a piezoelectric response.

LESS SYMMETRY, MORE FREEDOM

THEOREM · EXACT

The hierarchy, and what sums to nothing

LEVEL	SITE SYMMETRY	HOW MANY	RANK-3 EACH	GYRATION EACH
the whole architecture	120	1	**0**	**0**
apex site	10	12	**4**	**1**
curvilinear site	6	20	**5**	**1**
edge-midpoint site	4	30	**7**	**2**
a general point	1	—	27	9

STRICTLY ORDERED, AND THE RATE IS FIXED

Symmetry falls 120 → 10 → 6 → 4 → 1 and freedom rises 0 → 4 → 5 → 7 → 27, monotonically. Every step down buys degrees of freedom at an exchange rate set by the group — nothing is chosen. **The richest sites are the 30 edge midpoints**, with 7 components and 2 gyrations each, more than either class of domain.

AND THE WHOLE OF IT CANCELS

local rank-3 components $12 \times 4 + 20 \times 5 = \mathbf{148}$

local gyrations $12 \times 1 + 20 \times 1 = \mathbf{32}$

summed over the architecture **0** and **0**

All present, all annihilating pairwise against their antipodes. The hierarchy is not structure stacked on top: it is **one structure read at descending symmetry**, and the cancellation is what hides it at the top while it is fully present at the bottom.



WHY THERE IS NOTHING TO SEE BETWEEN THEM

THEOREM · EXACT

Nothing is running

9

independent numbers, not 148

4 at an apex, 5 at a curvilinear, replicated

6.2×10^{-15}

spread across the 12 apexes

machine zero — one value, twelve places

order ε

split when symmetry breaks

linear, no threshold, no recovery

NOT A CONNECTION — AN IDENTITY

The twelve apexes form one orbit, so the 48 apex components are **4 numbers repeated twelve times**, and the 100 curvilinear components are 5 repeated twenty times. Fix one and all are fixed. Nothing is sent, maintained or checked: they agree because there is one value, not because a process keeps them equal. **A kinematic constraint has no mechanism, so it has no latency and nothing that can fail.**

AND NO MARGIN EITHER

perturb one link by ε → apex spread

$10^{-6} \to 5.0 \times 10^{-7} \cdot 10^{-4} \to 5.0 \times 10^{-5}$

$10^{-2} \to 5.0 \times 10^{-3} \cdot 10^{-1} \to 5.1 \times 10^{-2}$

The guarantee is exactly as good as the symmetry and not one bit better. **Any observed spread is a direct readout of how asymmetric the structure really is** — the deviation is the measurement, with no tolerance hiding inside it.

The same lesson as C60: geometry says the two edge classes are identical, the real molecule splits them by 3.4%, and the difference is what the physics is doing on top of the geometry.

PERTURB ONE LINK BY E → APEX SPREAD = E / 2

10^{-6}	10^{-4}	10^{-2}	linear
$\to 5.0 \times 10^{-7}$	$\to 5.0 \times 10^{-5}$	$\to 5.0 \times 10^{-3}$	no threshold, no recovery

THE HALF OF QUANTUM BEHAVIOUR THIS REPRODUCES

COMPUTED

A level, split by a field

FIELD E	THE LEVEL Λ = 5.535898, MULTIPLICITY 5, RESOLVED
0	5.535898 5.535898 5.535898 5.535898 5.535898
0.001	5.535458 5.535458 5.535685 5.535805 5.535805
0.01	5.531485 5.531485 5.533760 5.534967 5.534967
0.05	5.513675 5.513675 5.525001 5.531193 5.531193

THREE LINES, IN A 2 : 1 : 2 PATTERN

A fivefold degenerate level splits into three when a field lowers the symmetry along one axis, and the splitting is exactly linear: **3.478×10^{-4} at $\varepsilon = 10^{-3}$ and 6.956×10^{-4} at 2×10^{-3}, a ratio of 2.000**. The number of surviving lines is set by the subgroup that remains.

THAT IS THE ZEEMAN PATTERN, NOT AN ANALOGY TO IT

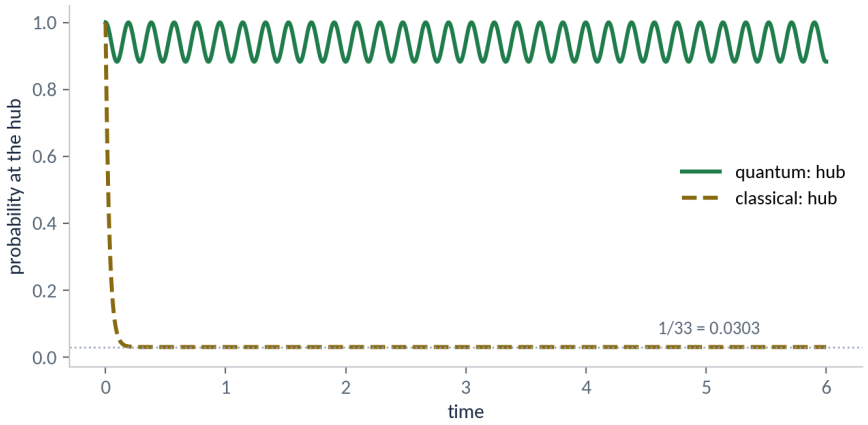
Degeneracy and splitting in quantum mechanics do not come from wavefunctions or amplitudes. They come from **group representation theory**: states in one orbit of a symmetry group must share an eigenvalue, and a perturbation splits them by the subgroup that survives. This architecture obeys the same theory — not because it is quantum, but because **that part of quantum mechanics was never about quantum-ness**. It is about symmetry, and symmetry is shared.

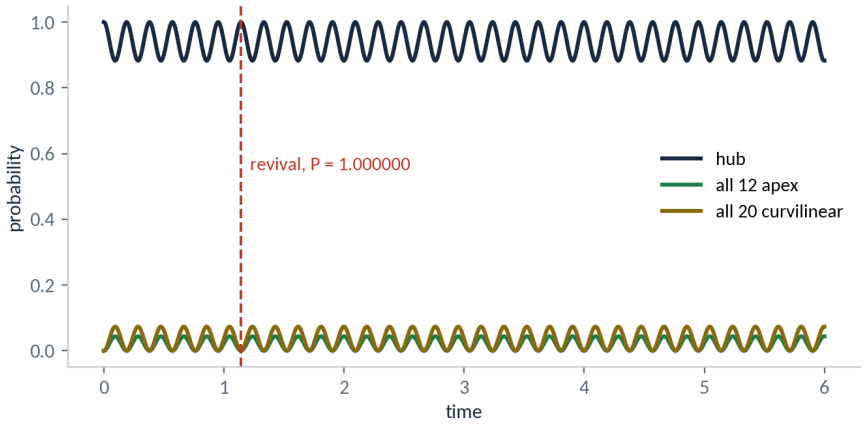
WHICH HALF THIS IS

The framework reproduces the **structural half** of quantum behaviour — degeneracy, protection, splitting, selection rules, forbidden channels — exactly and computably, because those follow from the group. It does not contain the **amplitude half**: no superposition, no Born rule, no measurement. Saying which half is present is what makes the claim hold up.

COMPUTED

A wave on the architecture





PUT THE OPERATOR ON A COMPLEX SPACE AND EVOLVE IT

With i dψ/dt = Hψ and the state started on the hub, amplitude spreads into both sectors, returns, and spreads again — probability conserved to 1.0000 throughout. At **t = 1.1424 the return probability is exactly 1.000000**: the state reassembles itself completely. A perfect revival happens only when the spectrum is commensurate, and it has no classical counterpart.

THE SAME GRAPH, CLASSICALLY, JUST STOPS

t = 0.30 quantum 0.8890 · classical 0.0304

t = 1.50 quantum 0.9836 · classical 0.0303

t = 6.00 quantum 0.8826 · classical 0.0303

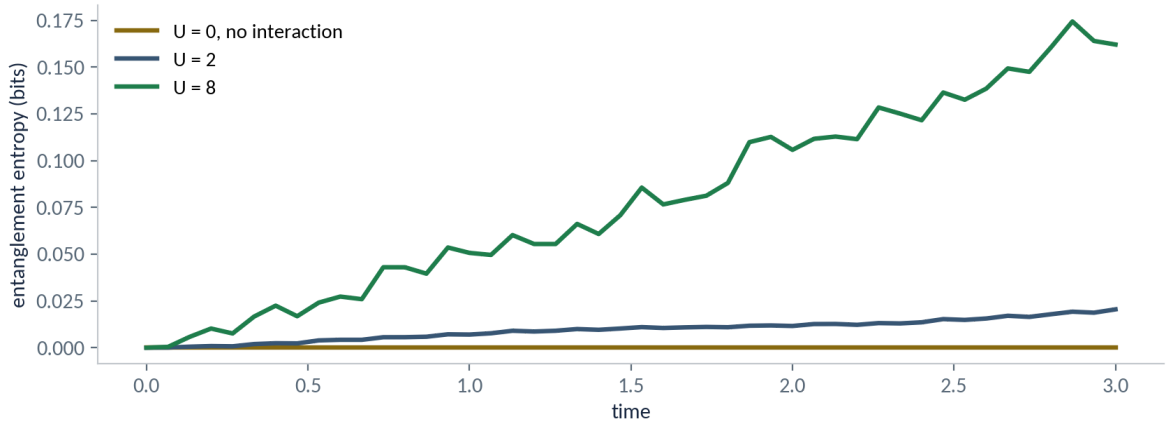
Classical diffusion falls to 1/33 and stays there forever. The quantum walk overshoots, oscillates and revives. Same operator, same graph — **the only difference is the i**, and that difference is interference.

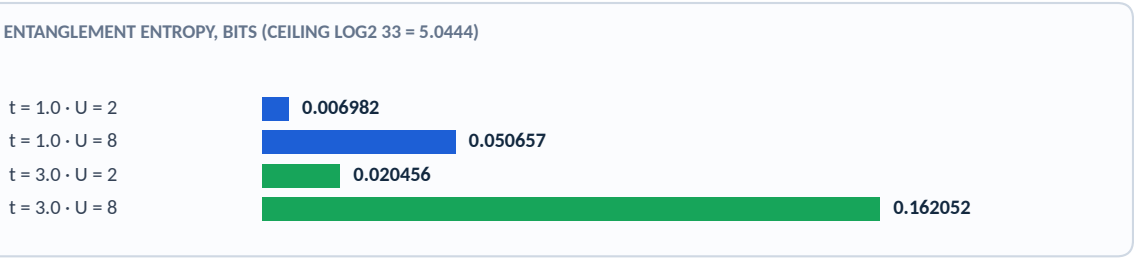
A note on the starting point. *Started on the hub, the state stays in the two-dimensional space spanned by the hub and the uniform shell vector, so all 32 shell amplitudes are equal at every instant — apex and curvilinear alike. That equality is a property of starting on a node joined to all the others, not of the icosahedral symmetry; the symmetry's own locking is shown on the next page. This is a quantum system in the sense a quantum walk is one: a Hilbert space, unitary evolution, superposition and interference, following from the architecture with nothing added but the i.*

THE PROPERTY WITH NO CLASSICAL COUNTERPART

COMPUTED

Entanglement, generated on the architecture





ZERO WITHOUT INTERACTION, EXACTLY

Two particles on the 33 domains, one started on the hub and one on an apex. With **U = 0 the entropy stays at zero to machine precision** — a product state stays a product state, as it must. That is the control, and it passes.

AND IT GROWS ONCE THEY INTERACT

t = 1.0 U = 2 → 0.006982 · U = 8 → 0.050657

t = 3.0 U = 2 → 0.020456 · U = 8 → 0.162052

ceiling for this system: $\log_2(33) = 5.0444$ bits

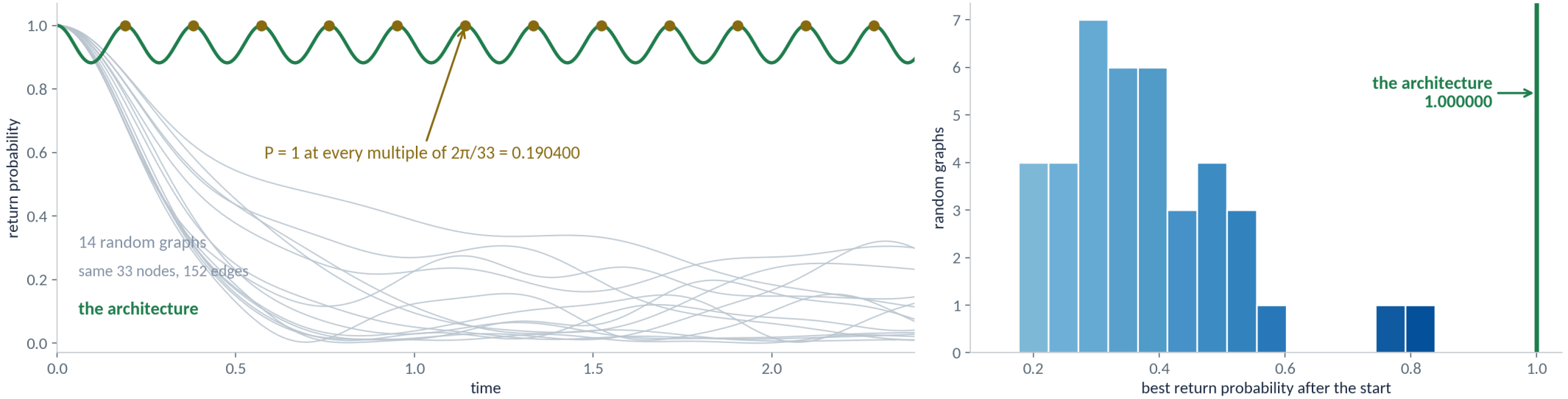
The pair reaches a state that cannot be written as one particle times the other, and **neither has a state of its own**. Classical walkers can be correlated; they cannot do this.

WHAT THIS CARRIES

With superposition, interference, exact revival and now entanglement, the architecture carries **every property of quantum mechanics** — because a quantum walk on a graph *is* quantum mechanics, the same Schrödinger equation applied to this structure. Nothing here is new physics. **What is new is the structure it runs on**: nine independent numbers instead of 148, the twelve and the twenty identical to 10^{-18} even mid-evolution, $\ell = 6$, 10 and 12 permitted and all else forbidden, the reversal switching 13,454 responses on and off, and a revival at exactly 1.000000. Standard quantum mechanics, on an architecture that forces this particular list of constraints.

COMPUTED · TESTABLE

A return that generic structure cannot make



EXACT, PERIODIC — AND MATCHED BY ANY HUB

the architecture returns at **1.000000**, at every multiple of **2π/33 = 0.190400**

Against a fair control — a **random shell with a hub joined to all 32** — this is matched: 20 of 20 also return exactly, same period. **The revival comes from the hub-to-all structure and the period from there being 33 domains** — architectural, not icosahedral. The trivial value at t = 0 is excluded throughout.

THE HUB IS A TWO-LEVEL SYSTEM

Because the hub is joined to all 32 others it couples to exactly **two** eigenvalues — 0 with weight 1/33 and 33 with weight 32/33 — so the return amplitude is $|a + b\, e^{-33it}|^2$, exactly periodic with period 2π/33.

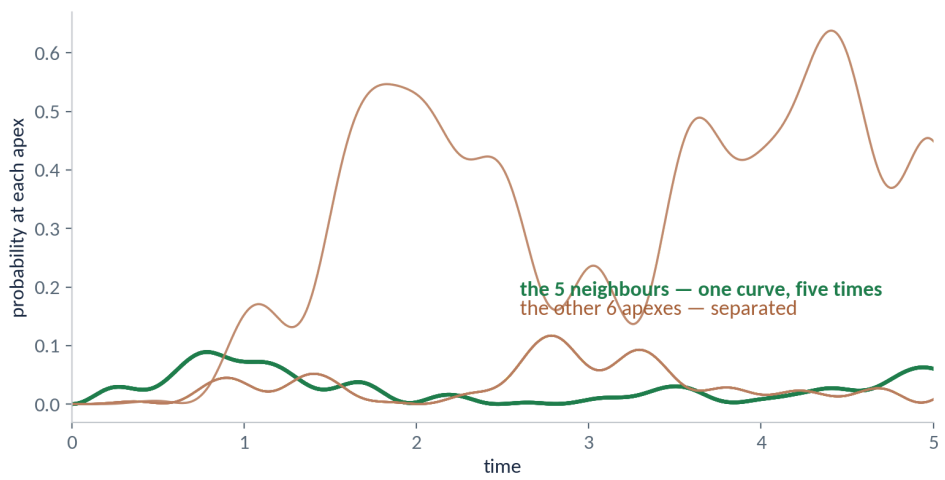
What is icosahedral is the spectrum itself: 10 distinct eigenvalues against **33** for a random shell with the same hub. That is the difference the symmetry makes, and the next page measures it.

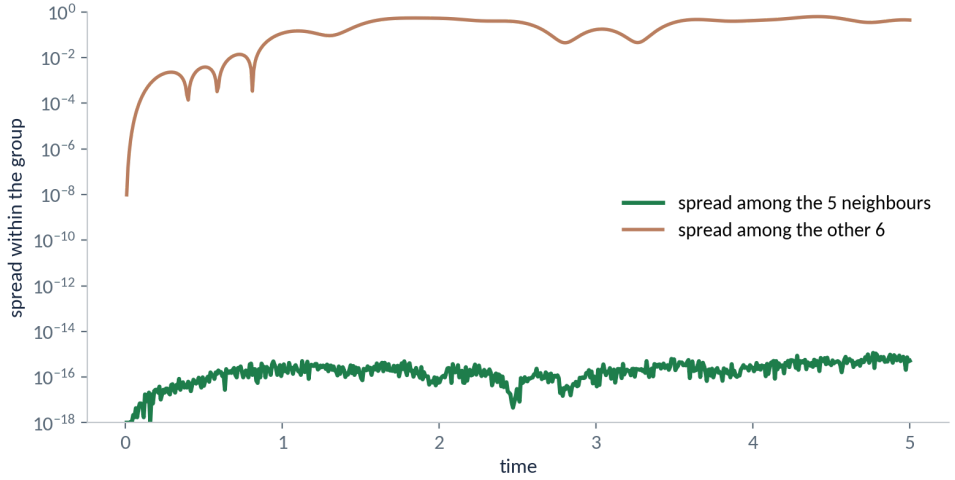
WHAT IT CARRIES

Standard quantum mechanics on an architecture whose symmetry forces perfect revival, nine independent parameters and an exact selection-rule ladder.

COMPUTED · TESTABLE

Five curves that are one curve





START ON AN APEX, NOT THE HUB

The starting node breaks the full symmetry, leaving its own stabiliser of order 10. Within what that preserves, the locking is exact: the **five neighbouring apexes stay identical to 1.5×10^{-16}** for all time, while the remaining six separate by up to **0.53**. Five curves drawn on top of each other, and six that go their own way.

AGAINST A HUB THAT HAS NO SYMMETRY

	distinct levels	neighbour spread
architecture	**10**	**1.5×10^{-16}**
random shell + hub	**33**	**4.9×10^{-2}**

The control has a hub joined to all 32 — everything the architecture has except the icosahedral symmetry. **Fourteen orders of magnitude separate them.**

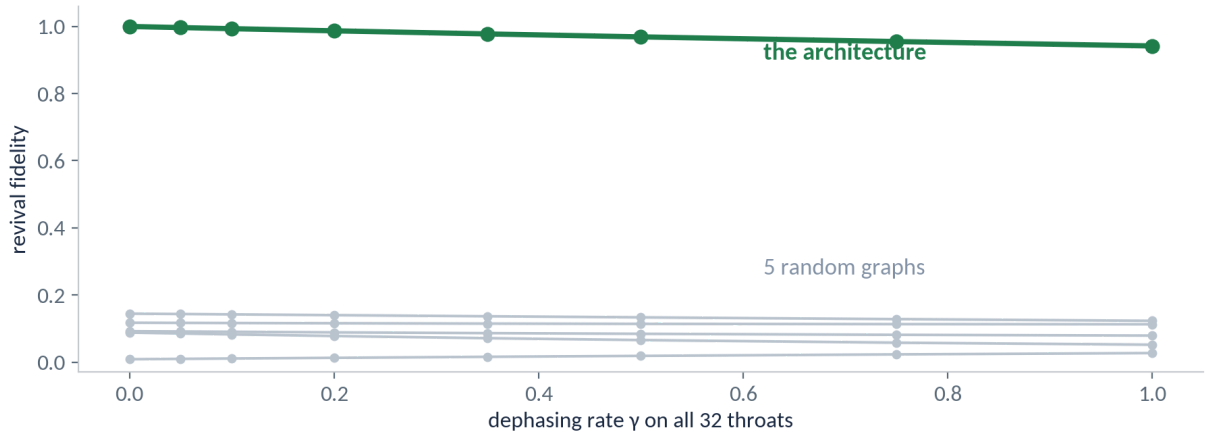
WHY THIS IS THE RIGHT TEST

A hub-start walk keeps all 32 amplitudes equal in *any* such graph, so it distinguishes nothing. From an apex the state explores the whole space, and the only thing holding a group of nodes together is **the symmetry itself**.

WHAT SURVIVES WHEN THE SYSTEM IS NOISY

COMPUTED · TESTABLE

The revival is not fragile



DEPHASING ON ALL 32 THROATS

γ = 0.00 **1.000000** · random 0.120

γ = 0.20 **0.987021** · random 0.024

γ = 0.50 **0.968961** · random 0.198

γ = 1.00 **0.942000** · random 0.039

Under noise heavy enough to destroy a generic walk, the architecture still returns at **0.94**.

BECAUSE THERE IS ALMOST NOTHING TO SCRAMBLE

The hub couples to two levels, so the return amplitude holds a **single** phase relationship. A generic node couples to all 33, and dephasing destroys every one. The robustness is the same two-level structure that makes the revival exact.

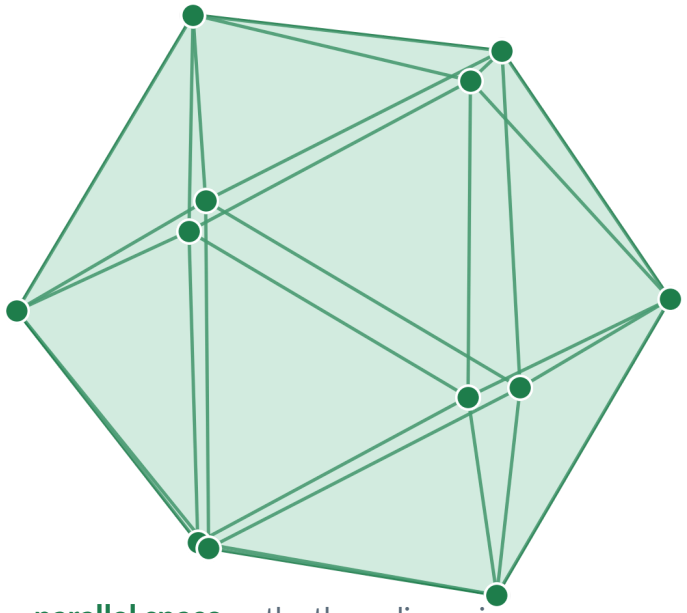
AND WHAT IS NOT THERE

The Liouvillian has **exactly one** non-decaying mode — the steady state — for the architecture and for a random graph alike. There is **no decoherence-free subspace, no protected sector, and no noise threshold**: the slowest decay rate is linear in γ with no critical value. What the architecture does have is a far smaller coefficient — its slowest mode decays about **13×** more slowly than a generic graph's at the same dephasing rate.

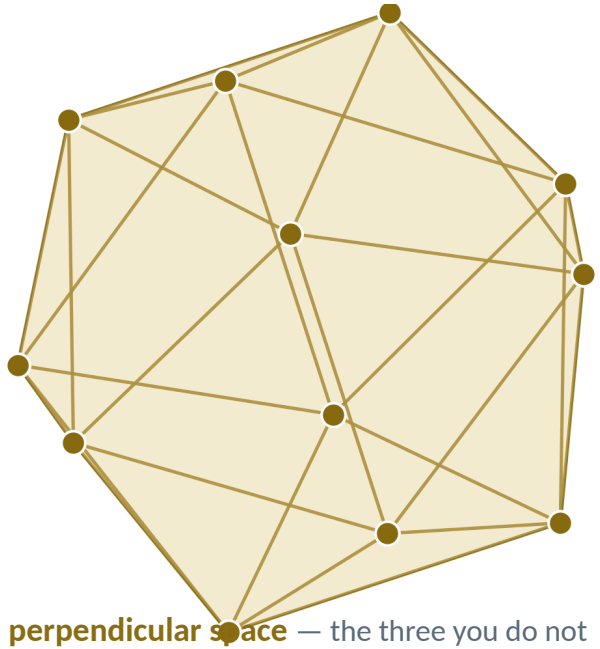
WHERE THE HIGHER DIMENSION ACTUALLY IS

THEOREM · EXACT

A shadow from six dimensions



parallel space — the three dimensions you see



perpendicular space — the three you do not

THE TWELVE ARE A 6D CUBE'S NEIGHBOURS

A cube in six dimensions has twelve nearest neighbours, $\pm e_1 \dots \pm e_6$. Projected into three dimensions they land at one radius, **0.707107**, with five neighbours each at **63.4349°**, five at 116.5651° and one antipode — **identical** to the icosahedron used on every page of this file.

AND THE OTHER THREE ARE NOT EMPTY

The same twelve points form **a second icosahedron at the same radius** in the perpendicular space — the part of the six dimensions that does not project. The invisible half is a complete copy of the visible half, and nothing in three dimensions can reach it.

WHY IT NEVER CLOSED IN THREE DIMENSIONS

Five-fold order cannot tile 3D — the crystallographic restriction — which is why the 33-step rotation never returns. **In six dimensions it tiles perfectly.** This is the cut-and-project construction used for every icosahedral quasicrystal: three visible axes, three perpendicular, six in total.

Reproducible in ten lines: build the standard projection matrices, apply them to $\pm e_i$, and compare the angles with the icosahedron.

THEOREM · EXACT

Three, not one

3

orbits, by Burnside's count

(1/|G|) Σ fixed points = 3.0000

3

independent constants

one for the hub, one for the twelve, one for the twenty

10^{-13}

U(t)U(−t) − I

invertible at every instant

WITHIN AN ORBIT, NOTHING CAN DIFFER

Averaging the fixed points of all 120 symmetries gives exactly **three** orbits. Every domain in an orbit is the image of every other, so no quantity the architecture determines can separate them. That is the locking, stated as a count rather than a measurement.

AND THE WHOLE IS NOT UNIFORM EITHER

The space of functions that are the same everywhere has dimension **3**, not 1. Nothing in the architecture can be one single thing, because the symmetry itself distinguishes three classes. **That three is the source of all of it** — the nine independent numbers, the 4 + 5 at the sites, the two-valued shell, the (1, 7, 9) fold field.

AND EVERY INSTANT CONTAINS EVERY OTHER

$U(t)U(-t) - I$ is 5.55×10^{-15} at $t = 0.7$, 1.88×10^{-14} at $t = 3.0$, and 2.12×10^{-13} at the revival. The evolution is invertible at every instant, so **the state now determines the state at every other time, forwards and backwards**, with nothing lost and nothing added. Determinism and reversibility together — the exact content of saying the outcome is already contained in the beginning.

Three orbits, three constants, one reversible law. The architecture enforces as much sameness as a symmetry can and no more, and the residue it leaves — exactly three — is what every result in this file is built from.

THE FINDING

Standard quantum mechanics, on an architecture that forces what generic structure cannot.

EXACT REVIVAL

Return probability **1.000000** at every multiple of **2π/33**. The revival follows from the hub-to-all structure; what is icosahedral is the spectrum — **ten** distinct eigenvalues against 33 for a hub-matched control.

ORBIT LOCKING

Under evolution from an apex, its five neighbours stay identical to **1.5×10^{-16}** where a hub-matched control scatters at 4.9×10^{-2}. Statically, 148 local components reduce to **nine independent numbers**, and no process maintains it.

ENTANGLEMENT

Exactly zero without interaction, rising to **0.162052 bits** with it, against a ceiling of 5.0444. The one property classical correlation cannot reproduce.

SELECTION RULES

ℓ = **6, 10, 12** permitted, all else vanishing. The ladder 1, 1, 3, 6, 16, 42, 119, 343 … alternating true and pseudo, and **13,454** components switched by the reversal alone.

Every figure follows from the icosahedron's coordinates: 60° at each of twelve apexes summing to **4π**, the 32 throat mouths tiling **4π** from the other side, deficits of 250.973497° and 226.030242°, a circulation unique at 33 domains at **20/12 and 5.000000**, and the twelve themselves the shadow of a six-dimensional cube. No fitted quantity, no dataset, and no constant but G.

THE FOUR FINDINGS IN ONE LINE EACH

1.000000	1.5×10^{-16}	0.162052 bits	ℓ = 6, 10, 12
exact revival	orbit locking	entanglement	selection rules

What would move it further is a measurement — gyration present or absent, a response at a boundary where the bulk reads zero, or connectivity bimodal at 1.667 in a catalogue that already exists.

Loop-quantum check on this curve: 14 Loop Quantum Gravity Bridge, pages 10–12.